

No. of Functional Features Included in the Solution

Date	03 July 2026
Team ID	PNT2026TMID-EG01
Project Name	EduGenie - AI-Powered Educational Assistant
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration & Login	Allows users to create an account and securely log in using email and password with JWT-based authentication.	auth.py, auth.html, login.html, register.html	Done	
2	AI Question Answering	Lets users ask any academic question and receive an accurate answer with educational context, powered by Gemini 1.5 Pro.	ai.py, gemini_service.py, ask-ai.html	Done	
3	Concept Explanation	Provides simplified explanations of complex topics for easier understanding by students.	ai.py, gemini_service.py, explain.html	Done	
4	AI Quiz Generation	Generates topic-specific multiple-choice quiz questions for self-evaluation, e.g. Pythagoras Theorem.	ai.py, schemas/quiz.py, quiz.html	Done	
5	Text Summarization	Condenses long study material or notes into concise, easy-to-review summaries.	ai.py, gemini_service.py, summarize.html	Done	
6	Personalized Learning Roadmap	Generates a structured beginner-to-advanced learning path for a chosen topic, e.g. SQL.	models/roadmap.py, ai.py, roadmap.html	Done	
7	Learning History Tracking	Stores and displays a user's past questions, explanations, and quiz attempts for progress tracking.	models/history.py, routers/history.py, history.html	Done	
8	User Dashboard & Profile	Displays an overview of user activity and allows users to view/update profile information.	routers/profile.py, dashboard.html, profile.html	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Registration & Login, AI Q&A, Quiz Generation, Summarization, Learning Roadmap)

Metric	Count / Value
Additional / Nice-to-Have Features	3 (Learning History Tracking, Dashboard, Profile Management)
Features Tested & Verified	8 of 8 (all features manually tested locally; see Performance Testing report)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	5	Dashboard, Quiz page, Roadmap page, Ask-AI page, Profile page
2	Backend / Logic	8	FastAPI routers (ai.py, auth.py, history.py, profile.py), Gemini service integration
3	Database / Storage	3	User model, History model, Roadmap model (SQLite via SQLAlchemy)
4	API / Integration	2	Google Gemini 1.5 Pro API, LaMini-Flan-T5 local model integration
5	Security / Authentication	1	JWT-based login/registration with password hashing